

Overview of Metadata Standards in Plant Sciences and Federated Discovery

Resources for Plant Sciences: Data Integration and Interpretation Tools

Tutorial Series · Systems biology, Multi-omics Integration and Modelling

GitHub: [NIB-SI/ECCB2026_T12](#) · 2026

Agenda

1. **Why metadata?**
2. **FAIR principles**
3. **Data standards**
4. **MIAPPE**
5. **Data repositories**
6. **Publishing workflows**
7. **Federated discovery**

The Problem: Data Without Context

Plant research generates enormous volumes of data
— but most of it cannot be reused:

- Measurements in lab notebooks or local spreadsheets
- No description of biological material, location, or conditions
- Variable names like `HT_1` , `yield_final` — meaningful only to the author
- No persistent identifier — impossible to cite, link, or find

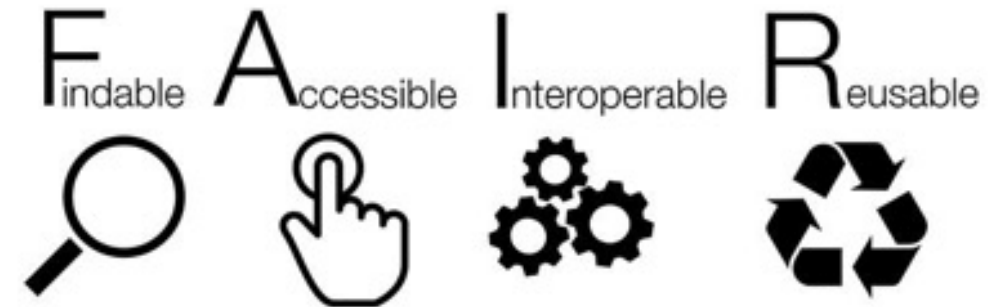
A dataset without metadata is just numbers



The FAIR Principles

Wilkinson et al. (2016) *Scientific Data*

	Principle	In practice
F	Findable	Persistent identifier (DOI), rich metadata, indexed
A	Accessible	Open protocol, authentication where needed
I	Interoperable	Controlled vocabularies, standard formats
R	Reusable	Clear licence, provenance, community standards



FAIR $\neq$ open access — data can be FAIR and still access-controlled

What This Session Covers

Why do metadata standards matter?

The plant sciences generate diverse, distributed data — phenotyping, genotyping, multi-omics. Without standards, this data cannot be reused, integrated, or discovered.

This session covers:

- The FAIR principles and what they mean for plant data
- MIAPPE — the plant phenotyping metadata standard
- Tools for publishing FAIR plant data (Dataverse, e!DAL-PGP)
- Federated discovery with FAIDARE

Why Standardise Phenotyping Data?

To enable reuse — by anyone, including yourself:

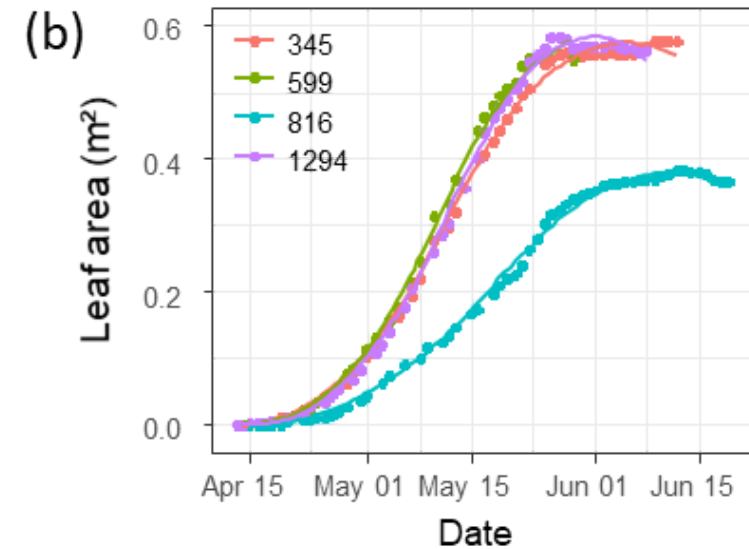
- Metadata about the experiment: who, what purpose, where, how

To enable automatic integration:

- Phenotype = measurement on a *cultivar* at GPS + time
- Genotype = marker alleles on a *cultivar*
- Climate = climatic data at GPS + time

To enable knowledge discovery:

- Metadata, controlled vocabularies, ontologies



Leaf area over time — same experiment, different genotypes

Phenotype Data Lifecycle

From data acquisition to knowledge

- **Raw data** — pheno/env measurements, variables
- **Derived data** — computed, reduced, indicators
- **Knowledge** — scientific publication and reuse

Each experimental dataset can generate **many derived datasets** for different scientific questions.

FAIR data enables **traceability, reproducibility & provenance** across the lifecycle.

Genotyping: Computed data +
Raw option

Phenotyping: Raw
measurements + Derived
indicators

**Solutions needed at every
stage:**

- Data standardisation
- Repositories for publication
- Findability / discovery

Data Standards for FAIR

Layer	Focus	Who drives it
Semantic	Controlled vocabularies, ontologies — describe the data	Biologist
Structural	Data models, formats — CSV, VCF, GFF, MIAPPE	Biologist + CS
Technical	Interoperability, tools — GA4GH, BrAPI	Computer scientist

Persistent Unique Identifiers underpin all layers: URI, gene ID, accession ID, trait ID, DOI

Metadata Standards in Plant Sciences

Domain	Standard	What it describes
Germplasm / biological material	MCPD — Multi-Crop Passport Descriptors	Species, accession, provenance, genebank ID
Phenotyping experiments	MIAPPE v1.1	Study design, variables, observation units
Environmental / metagenomic samples	MixS / MIMARKS (GSC)	Soil, environment, sequencing context
Trait definitions	Crop Ontology (CO)	Controlled vocabulary: trait + method + scale
Data exchange containers	ISA-Tab · RO-Crate	Structured archive packaging investigation + studies
Web services	BrAPI — Breeding API	REST API for breeding and phenotyping data
General dataset	Dublin Core · DataCite	Title, author, DOI — minimal discovery metadata

Most plant datasets need **more than one** — MIAPPE + MCPD + Crop Ontology is a typical combination

Data Repositories for Plant Data

Thematic repositories

- **Pros:** Precise dataset description, good quality metadata
- **Cons:** Not available for all data types (e.g. phenotyping); time-consuming curation

Examples: GnpIS, e!DAL-PGP, PIPPA, PHIS

Generalist repositories

- **Pros:** Any data type — flexible; rapid submission; author/title/DOI minimal metadata
- **Cons:** Weak biological description; non-standard formats; reduced FAIR compliance

Examples: Zenodo, Dataverse, Figshare

Generalist repos need **guidance + optional validators** to reach acceptable FAIR levels for plant data

Why Generic Repositories Need Companion Metadata

Generic repositories provide only broad metadata — year, data type, author, organism (no controlled vocabulary), keywords — and lack plant phenomics-specific fields:

- **Biological material** — from species to genetic resource accession
- **Traits** — observation variables with method and unit
- **Experiment locations** — GPS, environment conditions
- **Experimental factors** — treatments applied

Solution: dedicated metadata scheme as **companion files** — not filling long forms in the repository, but structured files alongside the data

This is exactly what **MIAPPE** provides.

MIAPPE

Minimum Information About Plant Phenotyping Experiments

What is MIAPPE?



MIAPPE v1.1 (Jan 2019) — the plant phenotyping metadata standard

- Crops and woody plants · open community: RFCs, GitHub, meetings & workgroups
- Stakeholders: ELIXIR, EMPHASIS, Bioversity, North American PPN

Papoutsoglou et al. (2020) Enabling reusability and interoperability of plant phenomic datasets with MIAPPE 1.1. *New Phytologist*, 227:260–273.

MIAPPE — Biologist Friendly

Minimal and sufficient metadata:

- Objective of the experiment
- Contributors
- Experimental procedures
- Biological material

Resources: clear definitions, examples, Excel templates, trainings

Collaborating with other standards:

- MCPD — biological material identification
- Crop Ontology — trait description
- Breeding API — web services
- ISA — data exchange container · RO-Crate

MIAPPE Implementations

Formats & APIs

- Ontology / OWL:
github.com/MIAPPE/MIAPPE-ontology
- File archive: ISA-Tab, RO-Crate
- Web services: Breeding API (BrAPI)

Databases (collecting)

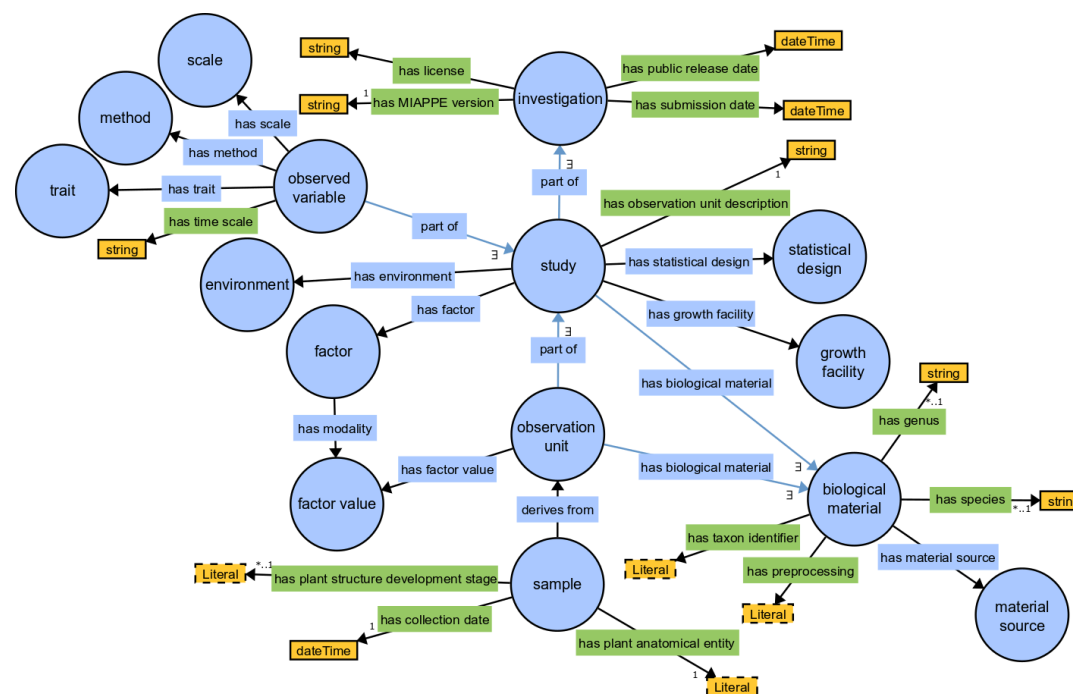
- PIPPA · PHIS

Databases (publishing)

- GnpIS · data.inrae.fr · e!DAL-PGP · BioSamples

Data portals

- rdmkit.elixir-europe.org — Plant Phenomics assembly
- www.miappe.org



MIAPPE data model — key entities and relationships

MIAPPE Structure

The ISA backbone and data model

MIAPPE v1.1 — The ISA Backbone

Level	Scope
Investigation	Whole dataset
Study	One experiment in one location, one or several years
Assay (Observed variable)	Trait or index (phenotypic or environmental)
Observation Unit	Plant · Microplot · Block · Trial...
Data file	Any format — tabular, images, NIR spectra...

Observed variable = **Level + Trait + Method + Scale/Unit**

MIAPPE v1.1 — Observed Variable (Assay)

Phenotyping/environment variable = Trait + Method + Unit/Scale

Example — Plant Height:

- **T1:** Plant Height
- **M1:** Total height · **M2:** First tassel branch · **M3:** Last expanded leaf · **M4:** Youngest growing leaf · **M5:** Highest pixel
- **U1:** cm · **U2:** mm · **U3:** pixel

Scope: plant-specific traits + general *-omics* (Gene Ontology)

MIAPPE v1.1 — Biological Material

Material Source — accession, cultivar/variety, region of provenance, laboratory cross

- MCPD: Genebank/Lab + Species + accession number · DOI

Biological Material — seed lot, cuttings, material used in the study

- Lab + internal accession number (mandatory) · URI · BioSample ID

Samples — detached leaves, other plant parts

- Lab + internal accession number · BioSample ID

MIAPPE v1.1 — Other Important Sections

Section	Content
Investigation	Title, description, contacts, licence
Study	Location, dates, experimental design
Assay	Observed variable definitions
Observation Unit	Hierarchy of experimental units
Sample	Physical samples taken
Experimental Factor	Treatments applied
Events	Irrigation, fertilisation, etc.
Environment	Climate, soil conditions
Files	Data files and their metadata

MIAPPE Specifications

Online resources

- Specification + checklist: www.miappe.org
- GitHub: github.com/MIAPPE/MIAPPE
- Ontology: github.com/MIAPPE/MIAPPE-ontology

Sections × metadata fields table with definitions, examples, cardinality, and links to controlled vocabularies

MIAPPE Check Item	Definition	Example	Format	Cardinality
0001 Investigation	Investigation is a research project that aims to answer a research question. It is a planned and systematic study that aims to answer a research question. It is a planned and systematic study that aims to answer a research question.	Investigation of the effect of a new drug on the treatment of a disease.	Text	1 per investigation
0002 Study	Study is a research project that aims to answer a research question. It is a planned and systematic study that aims to answer a research question. It is a planned and systematic study that aims to answer a research question.	Study of the effect of a new drug on the treatment of a disease.	Text	1 per investigation
0003 Person	Person is a human individual who is involved in the study. It is a human individual who is involved in the study. It is a human individual who is involved in the study.	Person who is involved in the study.	Text	1 per study
0004 Data File	Data File is a file that contains data. It is a file that contains data. It is a file that contains data.	Data File that contains data.	Text	1 per study
0005 Biological Material	Biological Material is a material that is used in the study. It is a material that is used in the study. It is a material that is used in the study.	Biological Material that is used in the study.	Text	1 per study
0006 Environment	Environment is a factor that influences the study. It is a factor that influences the study. It is a factor that influences the study.	Environment that influences the study.	Text	1 per study
0007 Experimental Factor	Experimental Factor is a factor that influences the study. It is a factor that influences the study. It is a factor that influences the study.	Experimental Factor that influences the study.	Text	1 per study
0008 Event	Event is a point in time when something happens. It is a point in time when something happens. It is a point in time when something happens.	Event that happens at a point in time.	Text	1 per study
0009 Sample	Sample is a group of individuals that are used in the study. It is a group of individuals that are used in the study. It is a group of individuals that are used in the study.	Sample that is used in the study.	Text	1 per study
0010 Observed Variable	Observed Variable is a variable that is measured in the study. It is a variable that is measured in the study. It is a variable that is measured in the study.	Observed Variable that is measured in the study.	Text	1 per study
0011 Variable ID	Variable ID is a unique identifier for a variable. It is a unique identifier for a variable. It is a unique identifier for a variable.	Variable ID that is used in the study.	Text	1
0012 Variable Label	Variable Label is a label for a variable. It is a label for a variable. It is a label for a variable.	Variable Label that is used in the study.	Text	1
0013 Variable Accession Number	Variable Accession Number is a unique identifier for a variable. It is a unique identifier for a variable. It is a unique identifier for a variable.	Variable Accession Number that is used in the study.	Text	1
0014 Trait	Trait is a characteristic of an organism. It is a characteristic of an organism. It is a characteristic of an organism.	Trait that is observed in an organism.	Text	1
0015 Trait Accession Number	Trait Accession Number is a unique identifier for a trait. It is a unique identifier for a trait. It is a unique identifier for a trait.	Trait Accession Number that is used in the study.	Text	1

Publishing Plant Phenotyping Data

Tools, repositories, and workflows

Plant Phenomics Tool Assembly

Data management planning

- Data Stewardship Wizard

Experimental data gathering

- MIAPPE Template
- File-based: FAIRDOM-SEEK, pISA-tree, COPO
- High-throughput: PHIS, PIPPA

Entry point & guidelines

- rdmkit.elixir-europe.org/plant_phenomics_assembly

Data sharing & repositories

- **Dataverses** — generalist, MIAPPE companion files
- **e!DAL-PGP** — plant genomics & phenomics, large data
- **Zenodo** — generalist, FAIR
- **GnplS** — plant genetics & phenomics (INRAE)

Metadata management

- MIAPPE ontology / ISA-Tab / BrAPI
- Frictionless validator (FAIRDOM)

FAIDARE — Federated Discovery Across Plant Databases

FAIR Data-finder for Agronomic REsearch

urgi.versailles.inra.fr/faidare/

A **federated search engine** indexing multiple plant science resources simultaneously — one query, many databases:

- Ensembl Plants · GnpIS · Gramene Plants
- SKM · KnetMiner · OPTIMAS-DW · brapi@EVA
- Filter by taxon group, data type, collection, database
- Ontology annotation search & expand

Data providers include: Gramene · EBI · INRAE-URGI · NIB · Rothamsted Research · EVA · IPK

Why FAIDARE matters:

Published data in MIAPPE-compliant repositories becomes **findable** through FAIDARE — closing the FAIR loop.

FAIR: Findable · Accessible
Interoperable · Reusable

FAIDARE = the **F** in FAIR for plant data

FAIDARE — Federated Discovery Across Plant Databases

Publishing with MIAPPE in Dataverse

Step-by-step process:

1. Create dataset in sub-dataverse
2. Add mandatory plant phenotyping metadata (MIAPPE fields)
3. Add generic metadata (title, authors, description)
4. Upload data files + MIAPPE companion file
5. Publish — get DOI

Good practice: include human-friendly provenance — links between files, README, variable definitions

Example: DROPS project dataset
doi.org/10.15454/IASSTN

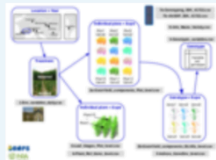
	10-Info-ObservedVariable.tab Tabular Data - 21.8 KB Published Nov 3, 2022 196 Downloads 17 Variables, 91 Observations UNF:6:bxOj...iaQ==  List of phenotypic and environmental variables used in the dataset, following the MIAPPE data standard (https://www.miappe.org/) Observed Variable
	11-Info-Study.tab Tabular Data - 6.4 KB Published Nov 3, 2022 180 Downloads 11 Variables, 19 Observations UNF:6:9nO7...GEQ==  List of studies, including locations, used in the dataset, following the MIAPPE data standard (https://www.miappe.org/) Study

MIAPPE companion files in a published
Dataverse dataset

Dataverse — Published Dataset Example

A multi-site experiment in a network of European fields for assessing the maize yield response to environmental scenarios

Version 4.0



Millet, Emilie J.; Pommier, Cyril; Buy, Mélanie; Nagel, Axel; Kruijer, Willem; Welz-Bolduan, Therese; Lopez, Jeremy; Richard, Cécile; Racz, Ferenc; Tanzi, Franco; Spitkot, Tamas; Canè, Maria-Angela; Negro, Sandra S.; Coupel-Ledru, Aude; Nicolas, Stéphane D.; Palaffre, Carine; Bauland, Cyril; Praud, Sébastien; Ranc, Nicolas; Presterl, Thomas; Bedo, Zoltan; Tuberosa, Roberto; Usadel, Björn; Charcosset, Alain; van Eeuwijk, Fred A.; Draye, Xavier; Tardieu, François; Welcker, Claude, 2019, "A multi-site experiment in a network of European fields for assessing the maize yield response to environmental scenarios", <https://doi.org/10.15454/IASSTN>, Recherche Data Gouv, V4, UNF:6:zS2/ccOQxFrKIUt+1S0Cvg== [fileUNF]

[Cite Dataset](#)

Learn about [Data Citation Standards](#).

Access Dataset

Contact Owner

Share

Dataset Metrics

15,218 Views

7,235 Downloads

3 Citations

Description This dataset comes from the European Union project DROPS (DROught-tolerant yielding PlantS). A panel of 256 maize hybrids was grown with two water regimes (irrigated or rainfed), in seven fields in 2012 and 2013, respectively, across along a climatic transect from western to eastern Europe, plus

DROPS project — multi-site maize experiment · doi.org/10.15454/IASSTN · [FAIR Cookbook](#)
[FCB083](#)

MIAPPE Template & Validation Tools

MIAPPE Template

- Excel/spreadsheet template for metadata collection
- MIAPPE v1.2 templates: github.com/MIAPPE/MIAPPE

Frictionless Validator

- Based on [Frictionless framework](#)
- Checks MIAPPE files for validity, tags Valid/Invalid
- Generates detailed validation reports
- Deployed as container, checks every 5 min

FAIRDOM-SEEK

- For confidential project data management
- ISA-based, MIAPPE compliant
- seek4science.org

Validator workflow:

1. Upload file to FAIRDOM
2. Wait 5–15 min for report
3. Review errors in report
4. Correct file in spreadsheet
5. Upload corrected version
6. Repeat until Valid ✓

Take-Home Messages

Entry points & guidelines

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 -
- [RDMKit Plant Phenomics assembly](#)
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Tools

Specialist repositories

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Acknowledgements

ELIXIR Plant Community & platforms

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